Muhammad Shaheer Luqman

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Applied AI & Software Engineer who ships end-to-end computer vision, NLP, and reinforcement learning systems for industrial clients — from research to full-stack delivery and cloud deployment. Published researcher (2× arXiv, IEEE), ICPC Gold Medalist, and summa cum laude graduate with a 3.90/4.00 CGPA.

Experience

Retrocausal *Remote (US-based)*

APPLIED RESEARCH ENGINEER *Jun 2025 – Present*

Building **AI systems** for industrial clients end-to-end, contributing to in-house frontend/backend software, and researching robot manipulation and Vision-Language-Action systems.

- Designed and shipped an end-to-end AI solution for **BMW**, owning frontend, backend, and AI components, cutting study compilation and runs from **2–3 hours of manual work to under 15 minutes**.
- Shipped in-house **full-stack** features across **Vue.js** frontend and **Django** backend, from API design to UI delivery.
- Deployed and scaled AI models on **AWS** and **RunPod** GPU infrastructure, achieving **~5x faster** step-name recommendation inference while **cutting GPU compute costs by up to 90%**.
- Conducted **research** on unsupervised action segmentation, temporal long-video understanding, robot manipulation, and reinforcement learning, **authoring two papers currently under review** (see *Research & Publications*).

Python | PyTorch | Computer Vision | VLMs | Full-Stack | Vue.js | Django | AWS | RunPod | Reinforcement Learning | Robotics

Confinality *Internship, Remote (Evening)*

MANAGEMENT TRAINEE OFFICER (MTO) — AI/NLP *Jul 2025 – Sep 2025*

Selected for the **MTO** trainee program, working on **NLP and speech-focused AI** workflows.

- Engineered **NLP** pipelines for **sentiment analysis, speech detection, and transcription**.
- Developed **voice generation** workflows reducing manual call handling for client operations.
- Iteratively improved our **AI calling agents**, raising their **call transfer rate from <1% to ~6%**.
- Delivered **call-performance analysis and next-step recommendations** to clients and internal teams.

NLP | Voice Generation | Sentiment Analysis | Speech Detection | Transcription | AI Calling Agents

Retrocausal *Internship, Remote (US-based)*

SOFTWARE ENGINEER INTERN *Mar 2024 – Jun 2025*

Worked on **industrial-scale AI systems** across computer vision, NLP, and LLM workflows.

- Built a video activity recognition pipeline using **Body Pose Estimation** and **Vision-Language Models** to automate PMTS code generation for industrial workflow optimization.
- Developed **LLM-based prompt engineering** systems for automated analysis of industrial process data.
- Generated automated **Excel reports** with graph and table visualizations, building **16 modules** in total.

Python | Django | Computer Vision | LLMs | Backend | MediaPipe | Excel Automation

Research & Publications

Enhancing Human-Likeness in Reinforcement Learning Agents via Hierarchical Macro Action Quantization 2026

arXiv:2605.30928

Under Review

A human-like RL framework that encodes human demonstrations into macro actions via a two-level hierarchical macro action quantization approach (HiMAQ), applied to dexterous control of a humanoid robot hand. On the D4RL benchmarks it outperforms the non-hierarchical MAQ baseline on human-likeness while matching or exceeding success rates, generalizing across IQL, SAC, and RLPD. *View on arXiv*

Unsupervised Skeleton-Based Action Segmentation via Hierarchical Spatiotemporal Vector Quantization 2026

arXiv:2604.15196

Under Review

A hierarchical spatiotemporal vector quantization framework for unsupervised skeleton-based temporal action segmentation, achieving SOTA on the HuGaDB, LARa, and BABEL benchmarks. *View on arXiv*

Ensuring Child Safety: An IoT-Based Surveillance System for Remote Monitoring and Detection of Anomalous Behavior 2024

ICETST 2024 • IEEE Xplore

Proposed an intelligent IoT-based system for remote child monitoring, combining hardware sensors, software components, and AI-driven anomaly detection to deliver real-time safety alerts. *View on IEEE Xplore*

Research Interests: Robotics, Computer Vision, IoT + AI Systems, Vision-Language Models.

Education

National University of Computer and Emerging Sciences (FAST) *Karachi, Pakistan*
BS COMPUTER SCIENCE *Sep 2021 – Jun 2025*

- CGPA: **3.90 / 4.00** (Ranked **9 / 440**)
- Summa Cum Laude Honors

Projects

PixelForge — AI-Driven Digital Marketing Platform
Final Year Project (GitHub)

- Built a full-stack generative AI platform using **React**, **Flask**, and **PostgreSQL** for automated advertisement image and video generation.
- Developed AI pipelines combining **BLIP** captioning, **Gemini**-powered slogan generation, and diffusion-based image synthesis for product-aware marketing content.
- Integrated **RunwayML Gen-1** workflows for AI-generated promotional videos using image-conditioned prompt engineering.
- Engineered an end-to-end system enabling users to generate branded advertisements from product images with minimal manual input.

LushWear — Business Operation Management
Client (GitHub)
Built, sold, and actively maintained for a local business to manage their day-to-day operations; currently in production use.

- Engineered scalable backend APIs using **FastAPI** for inventory, product catalog, stock tracking, and search operations.
- Built a cross-platform desktop application with **Electron**, featuring a modern responsive UI for inventory management workflows.
- Integrated **Supabase** for real-time database operations, authentication, and persistent cloud-backed storage.
- Built end-to-end **order management** with live order tracking via **courier APIs** and two-way **Shopify** integration for storefront sync.

Explore more projects on my *portfolio* and *GitHub*.

Skills

AI / Machine Learning PyTorch, Scikit-learn, Hugging Face; Computer Vision (YOLO, MediaPipe); Diffusion Models (Stable Diffusion); NLP (BERT, Whisper, ElevenLabs); Reinforcement Learning (IQL, SAC, RLPD); Robotics & Robot Manipulation; LLM Prompt Engineering.

Software Engineering Backend (Django, Flask, FastAPI, Express.js); Frontend (React, Next.js, Vue); REST APIs, GraphQL; Postman; Databases (MySQL, MongoDB, Oracle SQL, PostgreSQL); Cloud (AWS, RunPod); System Design.

Systems & Tools Selenium (Web Automation), Git & Version Control, Linux, Docker.

Programming Languages Python, C / C++, JavaScript.

Soft Skills Client-Facing & Stakeholder Communication, Attention to Detail, Ownership & Accountability

Honors & Awards

- **ICPC Gold Medalist** — ICPC National Competitive Programming *(Nov 2024)*
- **Summa Cum Laude** — Graduated with a high GPA (3.90 / 4.00), National University of Computer and Emerging Sciences (FAST) *(2025)*
- **ICPC Bronze Medalist** — ICPC National Competitive Programming *(Feb 2024)*
- **Winner** — Code Sprint, PROCOM 2023 *(2023)*
- **Best Team Award** — Coder's Cup, ACM FAST
- **4x Rector's List (4.0 GPA)** — NUCES (FAST), 4 semesters
- **4x Dean's List (3.5+ GPA)** — NUCES (FAST), 4 semesters

Leadership & Volunteer Experience

- **Director Technical Affairs** — Developers Day'25
- **Technical Lead** — ACM NUCES
- **CS Competition Head** — Developers Day'24
- **CS Competition Co-Head** — PROCOM'24
- **Training & Mentorship Head** — ACM NUCES

Certifications

IELTS Academic — English Proficiency Jul 2026
Overall Band: **7.5 / 9** • Reading **9.0** • Listening **8.5**